

Acero dúctil RDM 2019 - sección COL75X75FC28MPa

Project ID: Modelos_constitutivos

Case ID: COL75X75FC28MPa

Material: ductile_reinforcing_steel

Behavior: monotonic

Model ID: Mon_RDM2019

Source JSON: C:\Users\LENOVO\Documents\Proyectos python\StructureLab_PBD_RC\configs\stage_02\ductile_reinforcing_steel\monotonic\Mon_RDM2019.json

Response metrics

Point count: 801

Strain range: -0.1 to 0.1

Stress range [MPa]: -630.0 to 630.0

Reversals: 0

Failed points: 0

Calculated parameters

buckling_calculation_mode: rectangular_transverse_restraint

buckling_restraint_case: bending

epsilon_y: 0.0021

tie_area_mm2: 126.67686977437442

longitudinal_bar_inertia_mm4: 11976.69465197929

reduced_flexural_rigidity_N_mm2: 1227246004.377634

effective_restrained_bars: 2

bar_normalized_stiffness_N_per_mm: 119544.91776153668

tie_stiffness_N_per_mm: 126676.86977437443

equivalent_stiffness_ratio: 1.0596591820579464

buckling_intervals: 1

unsupported_length_mm: 100.0

s_over_db: 4.499437570303712

L_over_D: 4.499437570303712

rb: 9.221103051482205

buckling_active: False

residual_stress_mpa: 84.0

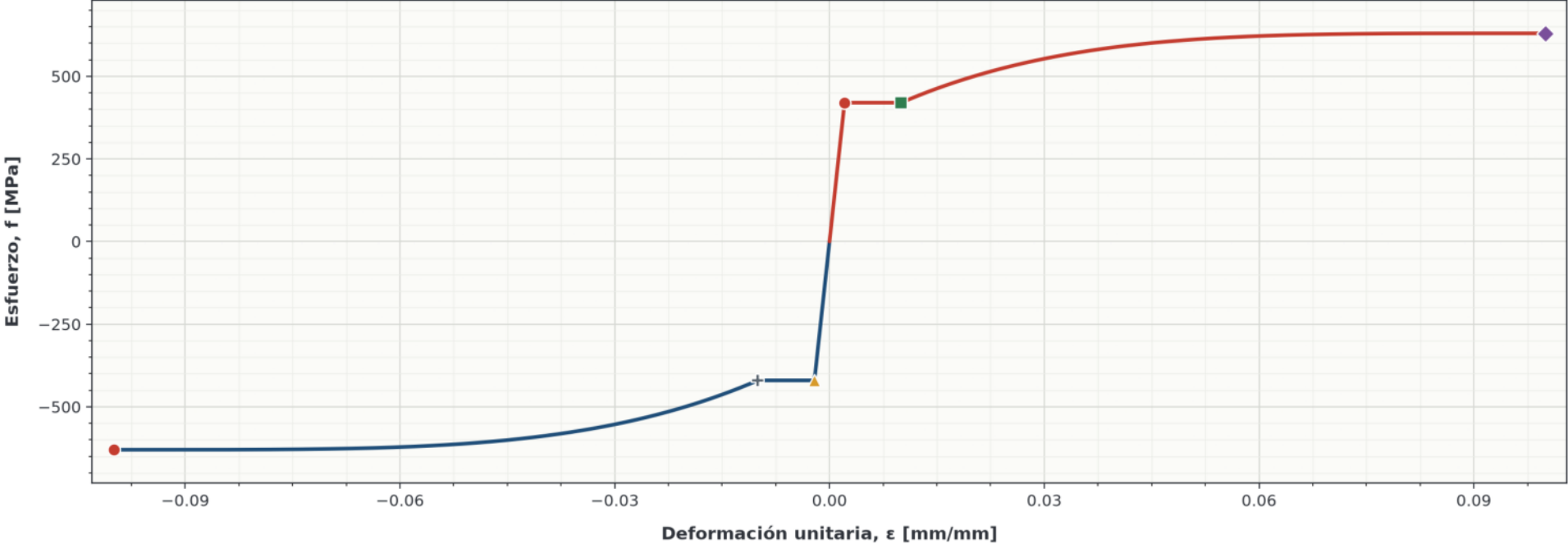
elastic_modulus_mpa: 200000.0

Warnings

None.

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Monotónico | Acero de refuerzo dúctil | Mon_RDM2019 | L/D = 4.499 | Pandeo inactivo | Puntos notables



Tracción

- Fluencia en tracción ($f_y = 420.000$ [MPa], $\epsilon_y = 0.002100$ [mm/mm])
- Inicio del endurecimiento en tracción ($f_{sh} = 420.000$ [MPa], $\epsilon_{sh} = 0.010000$ [mm/mm])
- ◆ Resistencia última en tracción ($f_{su} = 630.000$ [MPa], $\epsilon_{su} = 0.100000$ [mm/mm])

Compresión

- ▲ Fluencia en compresión ($-f_y = -420.000$ [MPa], $-\epsilon_y = -0.002100$ [mm/mm])
- + Inicio del endurecimiento de referencia en compresión ($-f_{sh} = -420.000$ [MPa], $-\epsilon_{sh} = -0.010000$ [mm/mm])
- Respuesta última en compresión ($f_{sc}(\epsilon_{su}) = -630.000$ [MPa], $-\epsilon_{su} = -0.100000$ [mm/mm])